Assignment 2

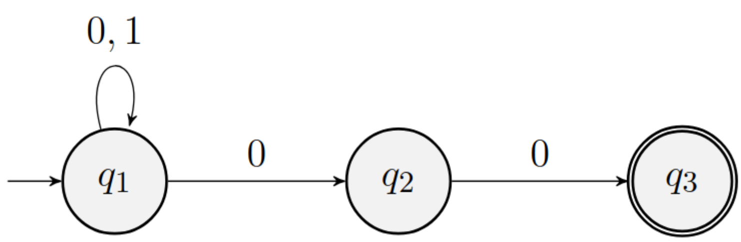
Note: there could be multiple versions of NFAs for the same language

Q1

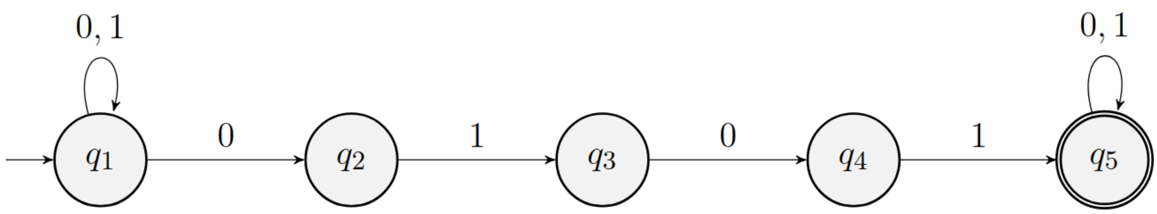
1) [0.20 points]
   Give state diagrams of the NFAs recognising the following languages over the following alphabet $\Sigma = \{0,1\}$:
   a. $\{w \mid w$ ends with $00\}$ using three states
   b. $\{w \mid w$ contains the substring $0101\}$ using five states
   c. $\{w \mid w$ contains an even number of 0s, or contains at exactly two 1s$\}$ using six states
   d. The language $\{0\}$ using two states
   e. The language $0^*1^*0^+$ using three states
   f. The language $1^*(001^+)^*$ with three states
   g. The language $\{\varepsilon\}$ with one state
   h. The language 0^* with one state

A1

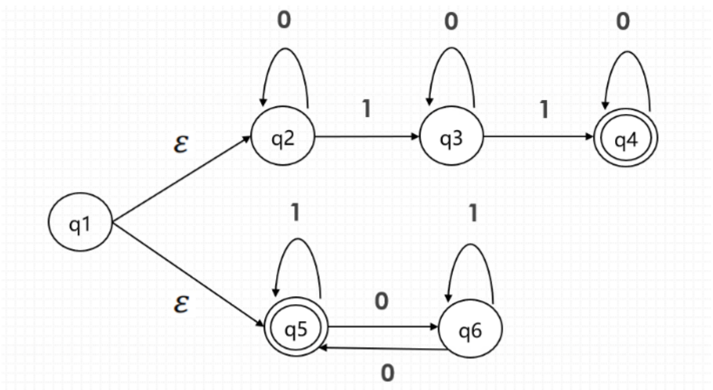
(a) a. $\{w \mid w$ ends with $00\}$ using three states



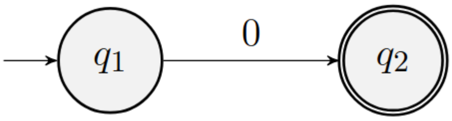
(b) b. $\{w \mid w$ contains the substring $0101\}$ using five states



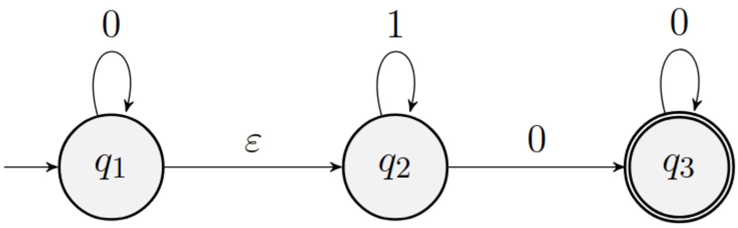
(c)
c. $\{w \mid w$ contains an even number of 0s, or contains at exactly two 1s$\}$ using six states



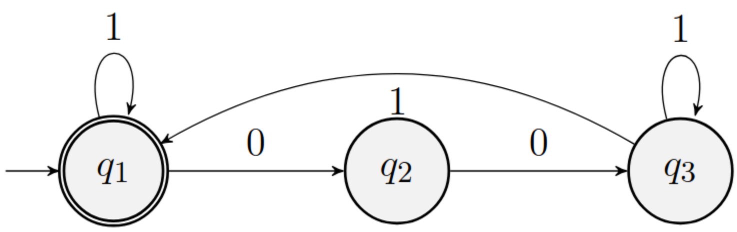
(d) d. The language $\{0\}$ using two states



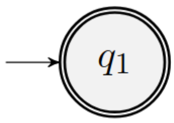
(e) e. The language $0^*1^*0^+$ using three states



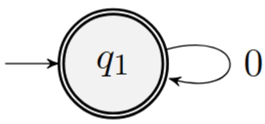
(f) f. The language $1^*(001^+)^*$ with three states



(g) g. The language $\{\varepsilon\}$ with one state



(h) h. The language 0^* with one state



Q2

2) [0.10 points]

Use the construction proof of closure under union of regular languages to give the state diagram of NFAs recognising the union of the following languages over $\Sigma = \{0,1\}$:

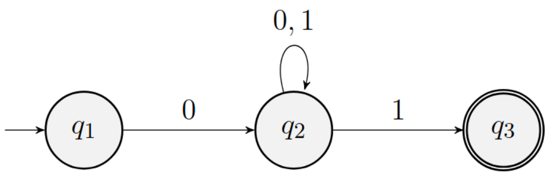
a. $L_1 = \{w|w$ begins with a 0 and ends with a 1$\}$
   $L_2 = \{w|w$ contains at least two 1s$\}$

b. $L_1 = \{w|w$ contains the substring 0101$\}$
   $L_2 = \{w|w$ does not contain the substring 111$\}$

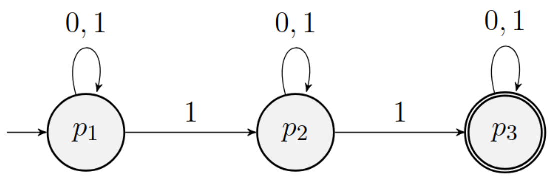
A2

(a) a. $L_1 = \{w|w$ begins with a 0 and ends with a 1$\}$
$L_2 = \{w|w$ contains at least two 1s$\}$

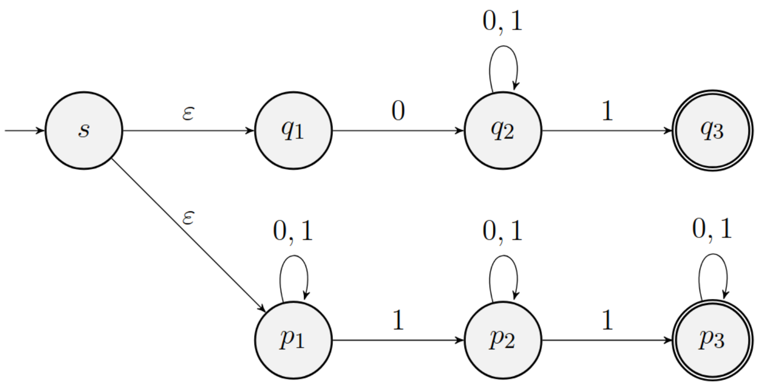
L1:



L2:



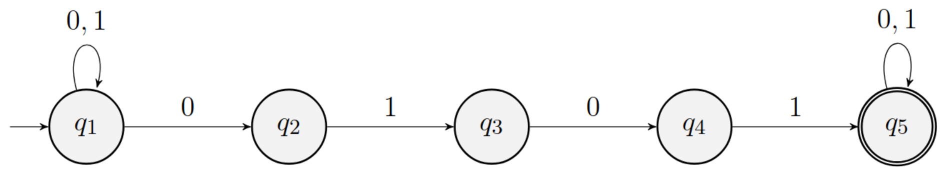
Result:



(b) b. $L_1 = \{w|w$ contains the substring 0101$\}$
$L_2 = \{w|w$ does not contain the substring 111$\}$

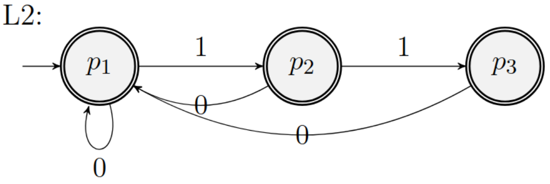
Version 1:

L1:



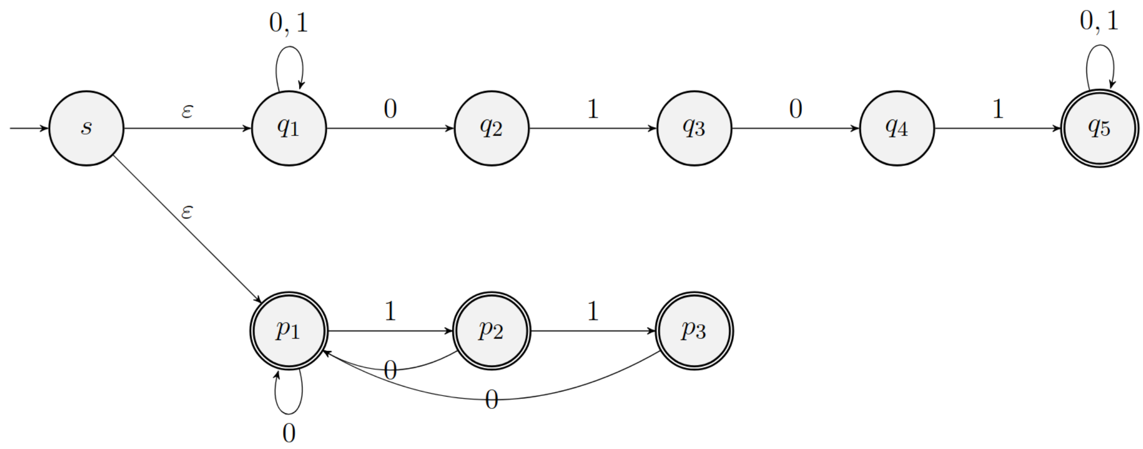
0, 1
0
1
0
1
0, 1
q_1
q_2
q_3
q_4
q_5

L2:



p_1
1
p_2
1
p_3
0
0
0

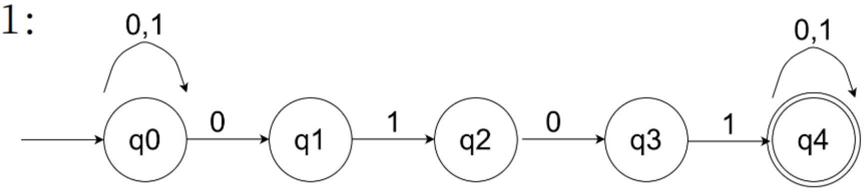
Result:



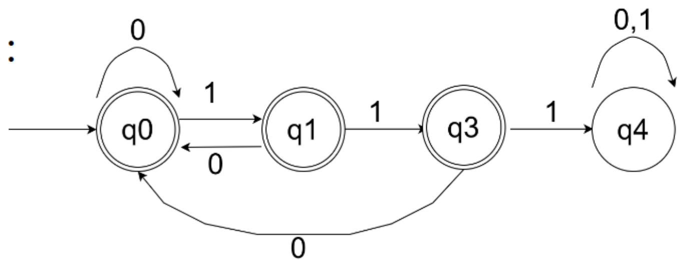
0, 1
0, 1
s
ε
q_1
0
q_2
1
q_3
0
q_4
1
q_5
ε
p_1
1
p_2
1
p_3
0
0
0

Version 2:

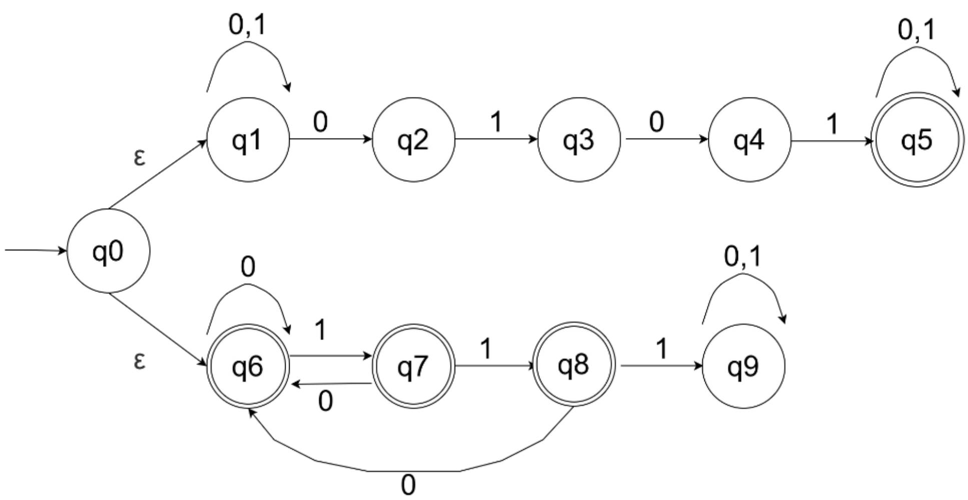
L1:



L2:



The Union of L1&L2



Q3

3) [0.10 points]

Use the construction proof of closure under concatenation of regular languages to give the state diagrams of NFAs recognising the concatenation of the following languages over $\Sigma = \{0,1\}$:

a. $L_1 = \{w|$ the length of w is at most 4$\}$
   $L_2 = \{w|$ every odd position of w is a 1$\}$

b. $L_1 = \{w|w$ contains at least four 1s$\}$
   The empty set

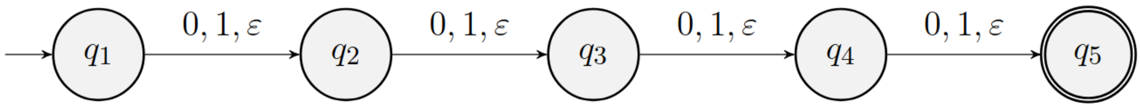
A3

(a) a. $L_1 = \{w |$ the length of w is at most 4$\}$

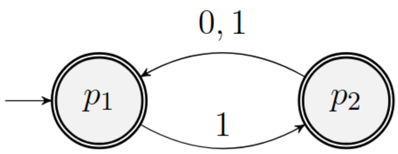
$L_2 = \{w |$ every odd position of w is a 1$\}$

Version 1

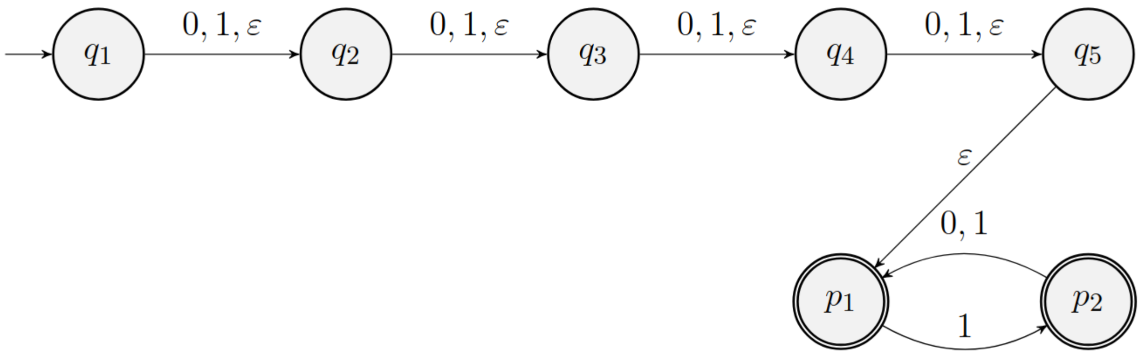
L1:



L2:

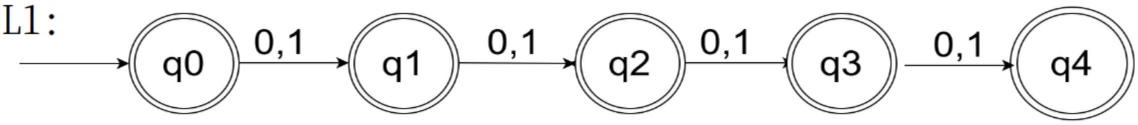


Result:

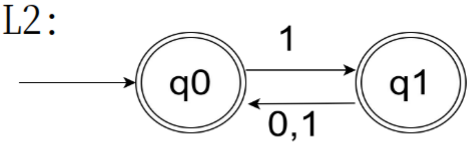


Version 2

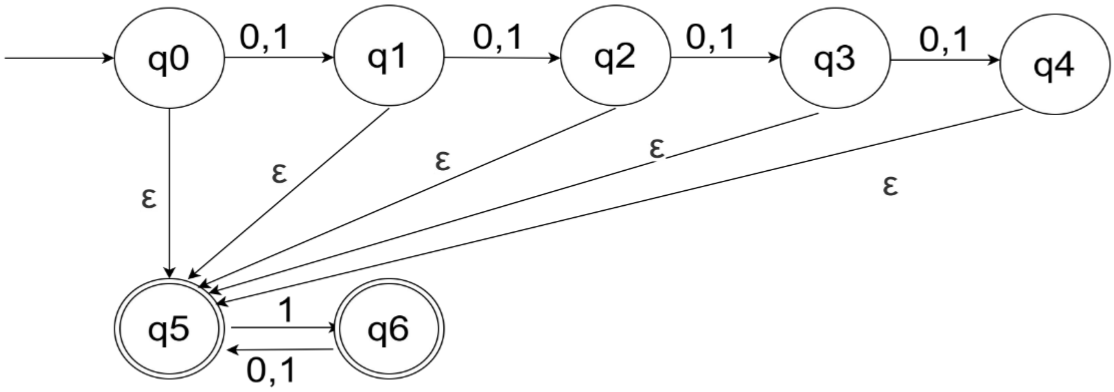
L1:



L2:



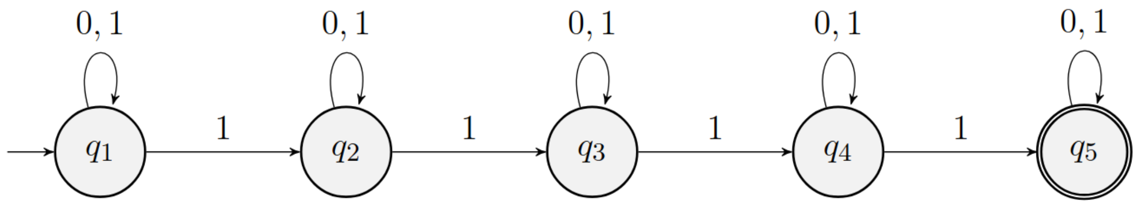
The concatenation of L1&L2:



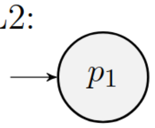
(b) b. $L_1 = \{w|w$ contains at least four 1s$\}$

The empty set

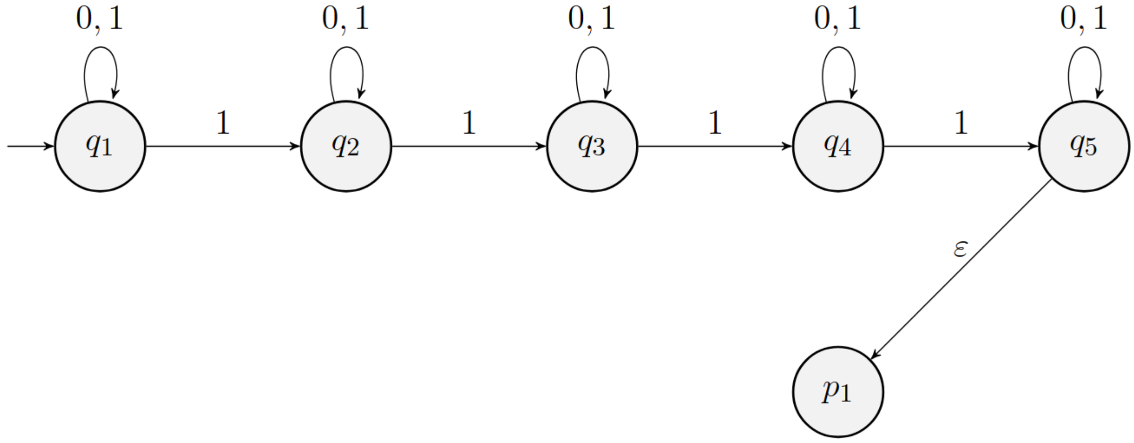
L1:



L2:



Result:



Q4

4) [0.15 points]

Use the construction proof of closure under Star of regular languages to give the state diagram of NFAs recognising the Star of the following languages over over $\Sigma = \{0,1\}$:

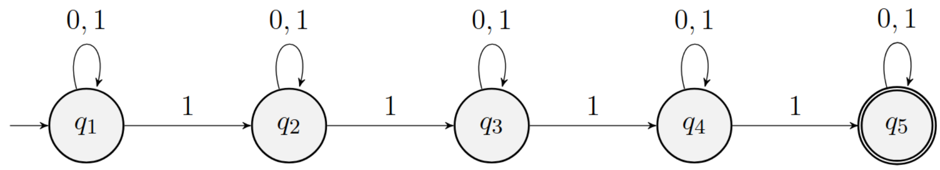
a. $L_1 = \{w|w$ contains at least four 1s$\}$

b. $L_1 = \{w|w$ contains at least two 1s and at most one 0$\}$

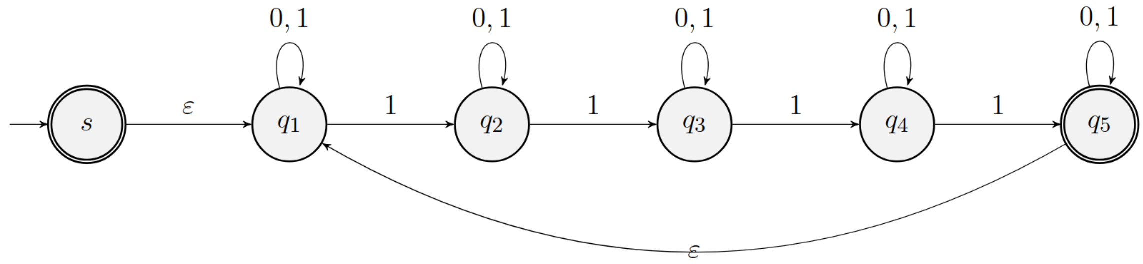
c. The empty set

A4

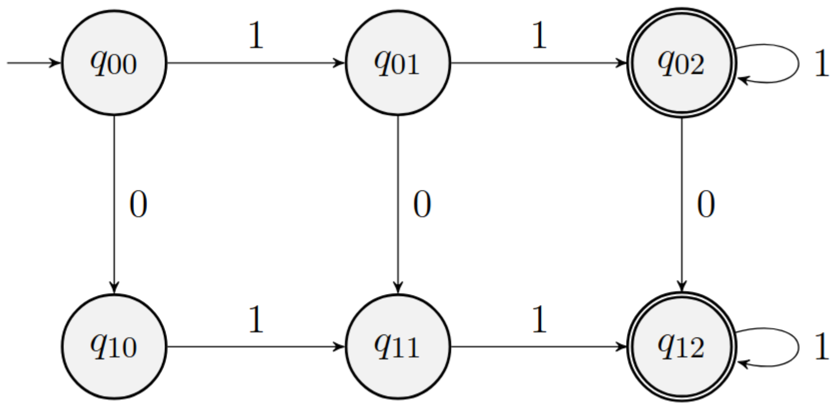
(a) a. $L_1 = \{w|w$ contains at least four 1s$\}$



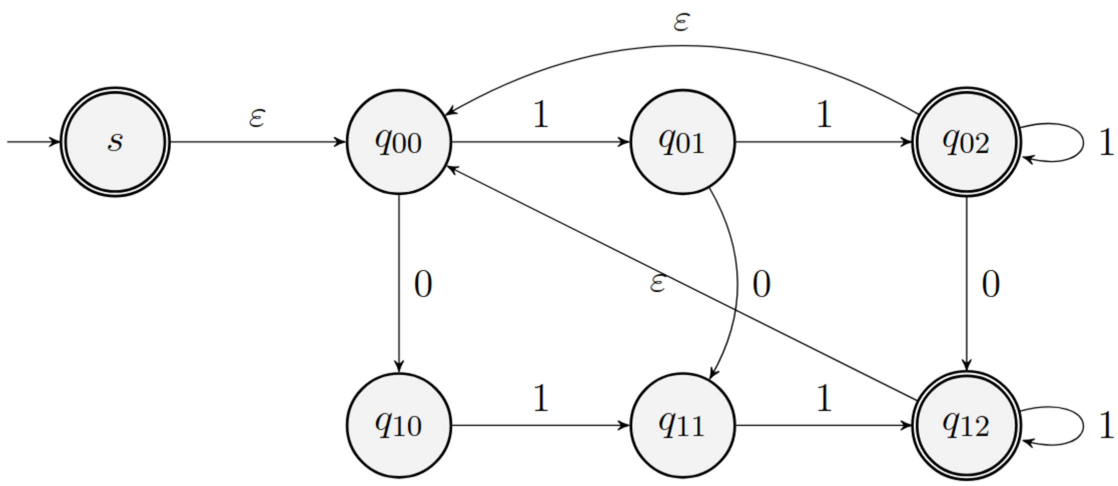
Result:



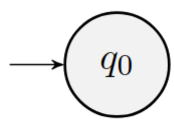
(b) b. $L_1 = \{w|w$ contains at least two 1s and at most one 0$\}$



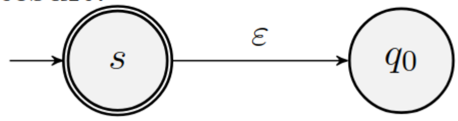
Result:



(c) c. The empty set



Result:



Q5

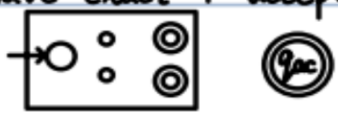
5)[0.05 points]

Prove that every NFA can be converted into an equivalent one that has a single accept state.

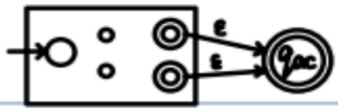
A5

5. Every NFA can be modified to have exact 1 accept state :

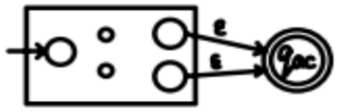
① Add a new accept state q_{ac}



② Add ε-paths from F to q_{ac}



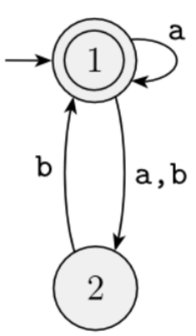
③ Leave q_{ac} only in F



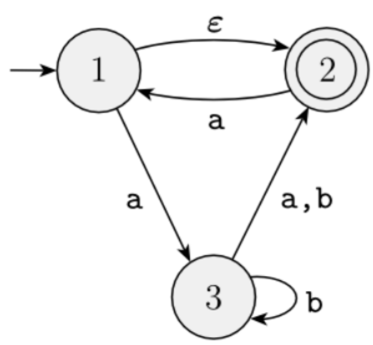
Q6

6) [0.40 points]

Convert the following NFAs into DFAs using the proof construction.



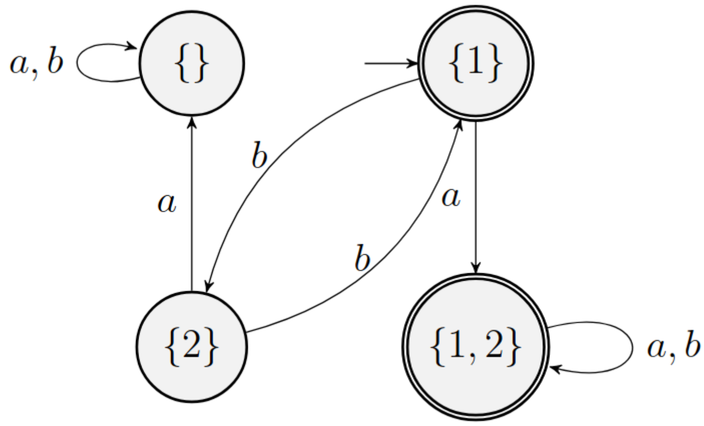
(a)



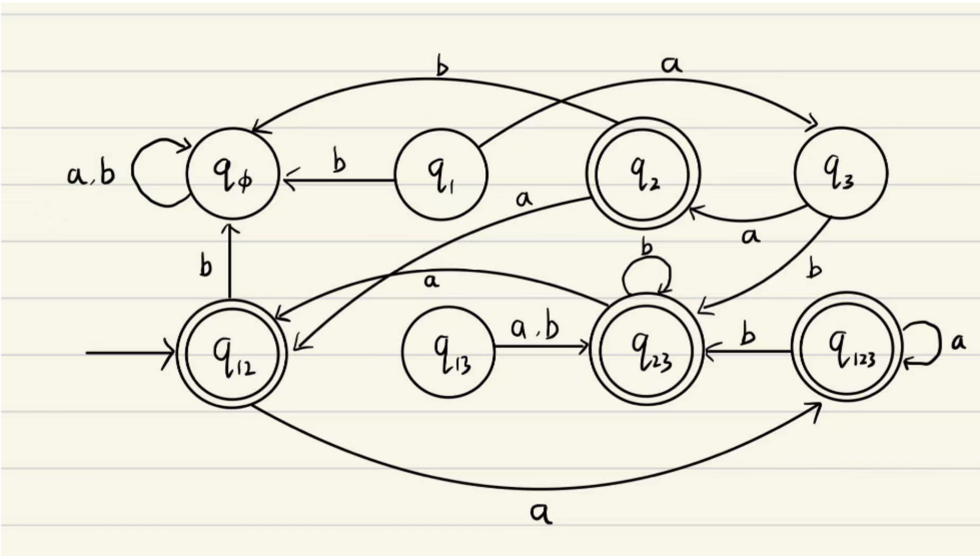
(b)

A6

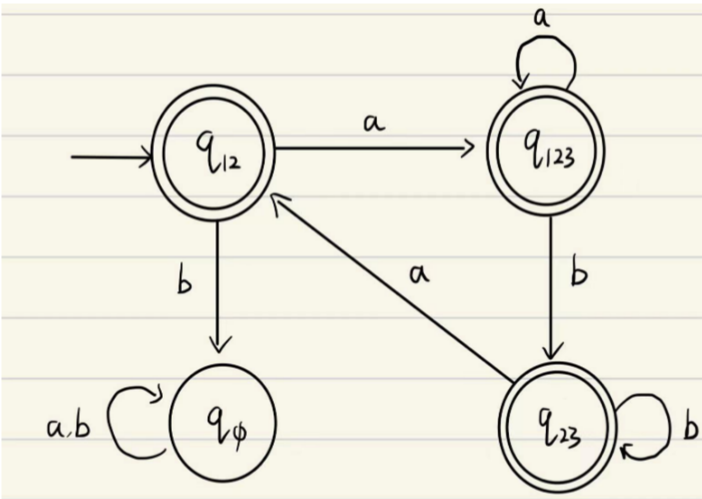
(a)



(b)



After Simplify:



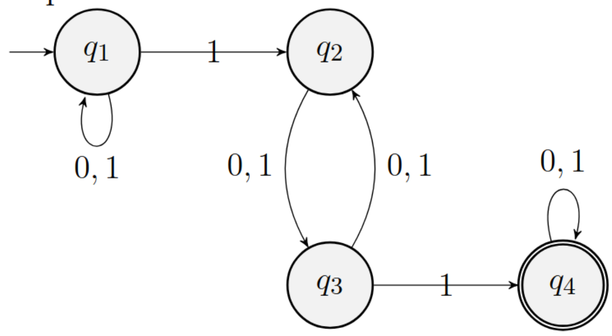
Q7

7) [1 point]

Let $F = \{w \mid w$ does not contain a pair of 1s that are separated by an odd number of symbols$\}$ over $\Sigma = \{0,1\}$. Give a state diagram of a DFA with five states that recognises F. (You might find it helpful first to find a 4-state NFA for the complement of F)

A7

complement of F :



By turn NFA to DFA and get of simple DFA of complement, we can find F :